EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2018
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME						
CIVICS GROUP	1	7	-			
REGISTRATION NUMBER						

H2 Biology

9744/03

Paper 3 Long Structured and Free Response Questions

18 September 2018

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on separate writing paper.
Additional Materials: Writing Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.
Tie your answers for Section B together using the string provided.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, submit section A and B separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4 or 5	
Total	**75**

This document consists of **15** printed pages and **1** blank page.

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For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Human diseases can be caused by many different types of organisms, such as bacteria and viruses.

(a) State two differences between the genetic material of bacteria and viruses.

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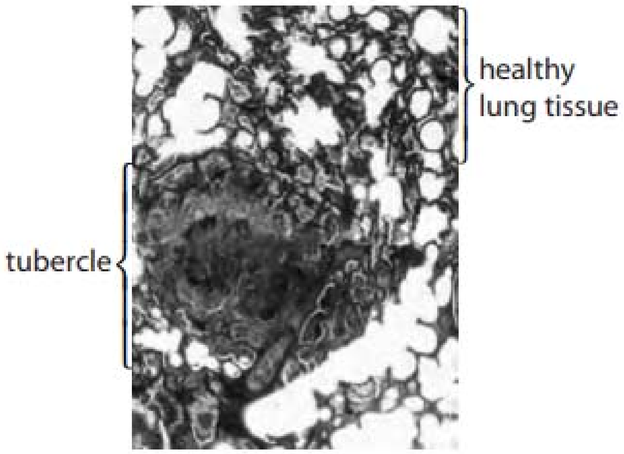
Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis*. Fig. 1.1 shows a tubercle in part of a lung infected by the bacteria.



Magnification ×50

Fig. 1.1

(b) Describe the role of the tubercle in transmission of tuberculosis to non-infected individuals.

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For Examiner's Use

(c) Mycolic acids are substances that form part of the cell wall of *Mycobacterium tuberculosis*. Isoniazid (INH) is an antibiotic that is used to treat tuberculosis. Fig. 1.2 shows the mode of action of isoniazid. Inactive isoniazid must be activated by a bacterial enzyme, **B,** to an active form.

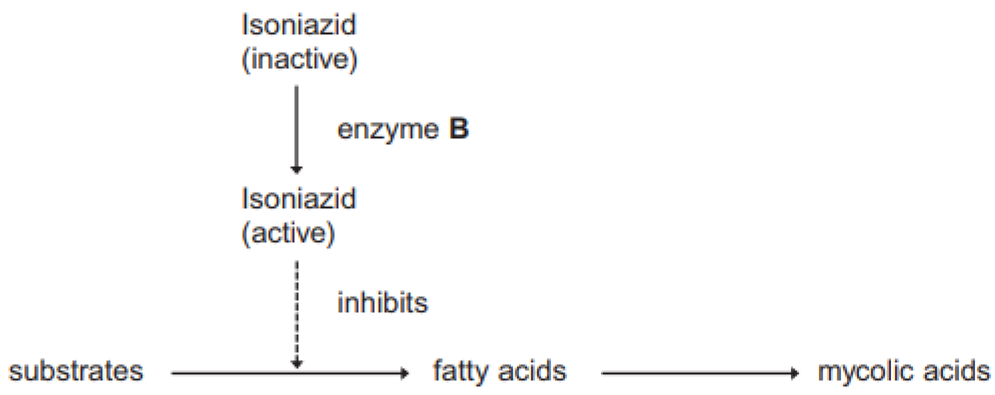

Fig. 1.2

(i) Explain the meaning of the term antibiotic.

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(ii) Explain the effect of active isoniazid on *Mycobacterium tuberculosis*.

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(d) Treating *Mycobacterium tuberculosis* infections can be a problem, as the bacteria are resistant to many antibiotics.

(i) Explain how mutation can result in resistance to isoniazid in *Mycobacterium tuberculosis*.

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(ii) Describe one way in which genes for antibiotic resistance are transferred horizontally between bacteria.

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(iii) The percentage of strains of *Mycobacterium tuberculosis* resistant to the antibiotic INH has increased from 2006 to 2008.

Explain how natural selection could have resulted in this increase.

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The ingestion of antibiotics can also have effects on the gut flora in an individual. Gut flora refers to the community of microorganisms in an individual's gastrointestinal tract. Fig. 1. 3 shows the diversity of the gut flora of an individual over the course of 18 months after taking a course of antibiotics for seven days.

The percentage of each type of bacteria in the gut flora was recorded for a period of 18 months.

Each type of bacteria is represented by a different letter in Fig. 1.3.

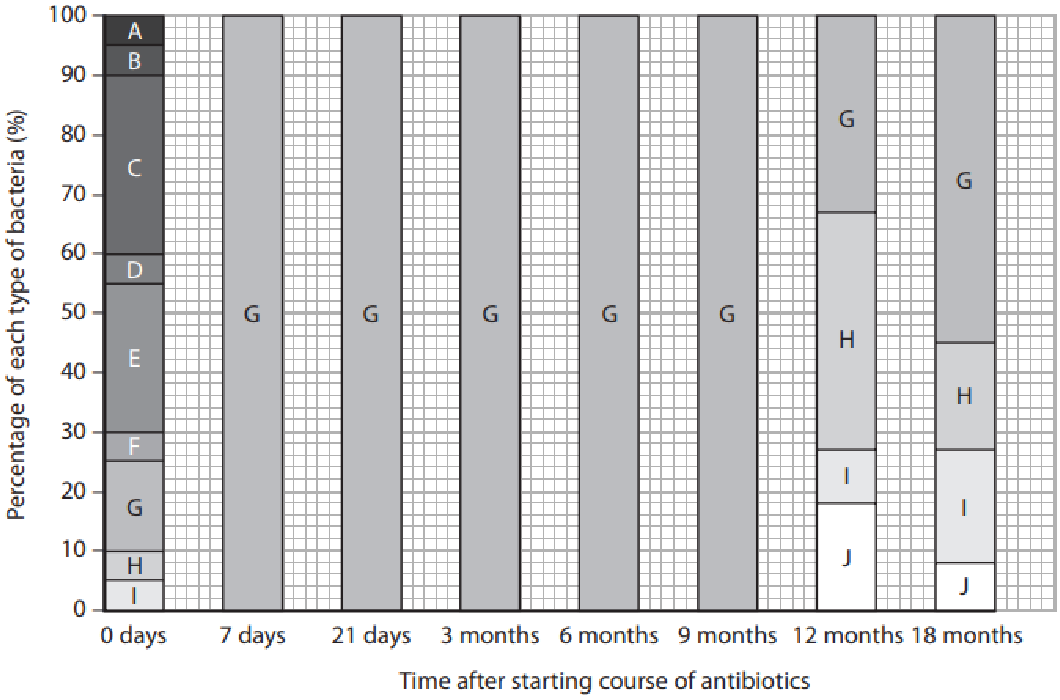

Fig. 1.3

(e) Using Fig. 1.3, describe the effect of this course of antibiotics on the diversity of gut flora in the individual.

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(f) *Mycobacterium tuberculosis* faces various stressful conditions inside the host, such as changes in pH, exposure to reactive oxygen species and degradative enzymes. The bacterium responds to such conditions through expression of the genes in the *mymA* operon.

The seven genes in the *mymA* operon, *RV3083* to *RV3089*, are involved in the modification of fatty acids required for the cell wall of pathogenic mycobacteria, and plays an important role in their virulence.

Fig. 1.4 shows the *mymA* operon.

Promoter	Operator	*RV3083*	*RV3084*	*RV3085*	*RV3086*	*RV3087*	*RV3088*	*RV3089*

Fig. 1.4

(i) Compare *mymA* operon and *trp* operon.

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(ii) A drop in pH results in the expression of *mymA* operon and another gene in *Mycobacteirum tuberculosis* which encodes a transcriptional regulator, VirS. VirS belongs to the AraC family and is involved in regulating the expression of the *mymA* operon.

Suggest how VirS regulates the expression of the *mymA* operon.

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(iii) Unlike the wild type bacteria, a mutated *Mycobacterium tuberculosis* has one of the structural genes in the *mymA* operon deleted.

Explain if the gene products coded for by the remaining structural genes in the *mymA* operon of the mutated bacteria are functional.

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(iv) In another mycobacterium, the operator is deleted.

State the effect of this deletion mutation on the expression of the *mymA* operon.

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[Total: 30]

For Examiner's Use

2 Huntington's disease (HD) is a genetic condition that leads to a loss in brain function. The gene involved contains a section of DNA with many repeats of the base sequence CAG. The number of these repeats determines whether or not an allele of this gene will cause Huntington's disease, as illustrated in Fig. 2.1 below.

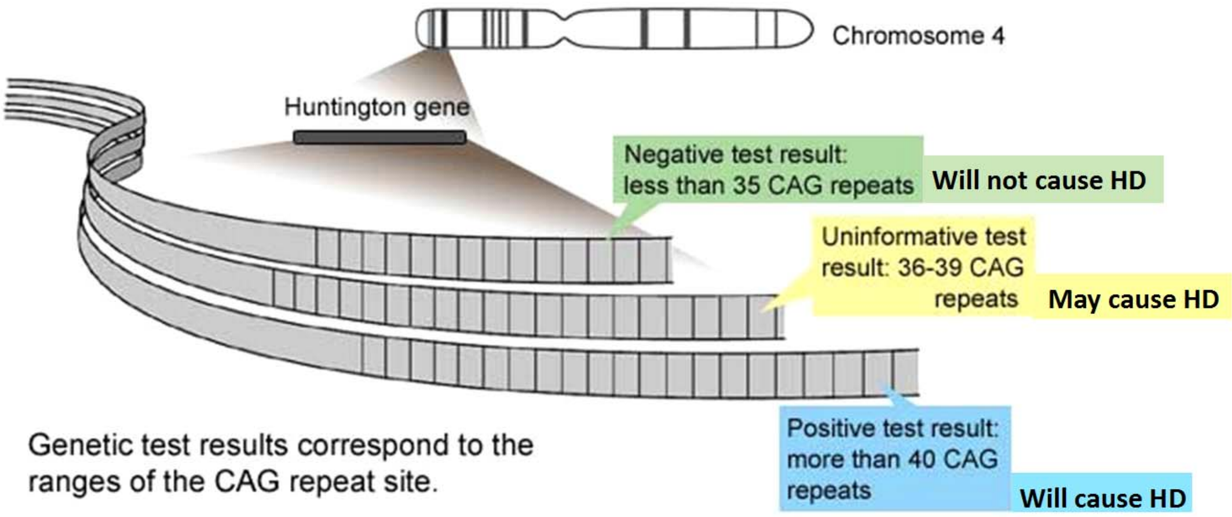

Fig. 2.1

Fig. 2.2 below shows the age at which each patient with HD first developed symptoms and the number of CAG repeats in the allele causing HD disease for each of these patients.

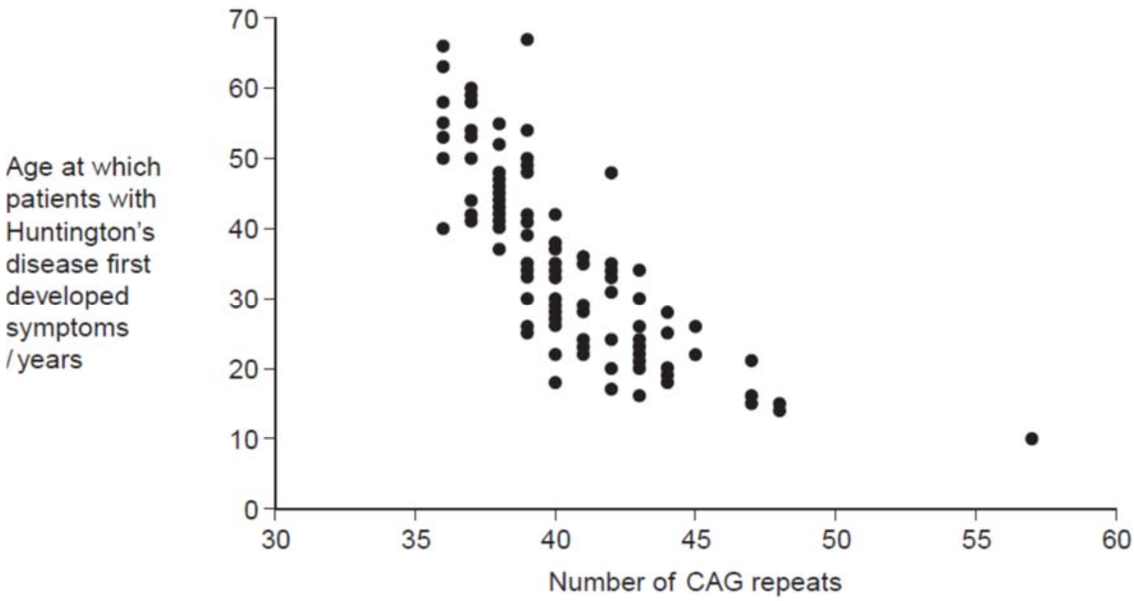

Fig. 2.2

(a) People can be tested to see whether they have an allele for this gene with more than 36 CAG repeats. Some doctors suggest that the results can be used to predict the age at which someone will develop HD.

(i) Use information in Fig. 2.2 to evaluate this suggestion.

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(ii) Huntington's disease is always fatal. Despite this, the allele is passed on in human populations. Use information in Fig. 2.2 to suggest why.

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(b) Scientists took DNA samples from three people, **J**, **K** and **L**. They used the polymerase chain reaction (PCR) to produce many copies of the piece of DNA containing the CAG repeats obtained from each person. They separated the DNA fragments by gel electrophoresis. A radioactively labelled probe was then used to detect the fragments. Fig. 2.3 shows the appearance of part of the gel after an X-ray was taken. The bands show the DNA fragments that contain the CAG repeats.

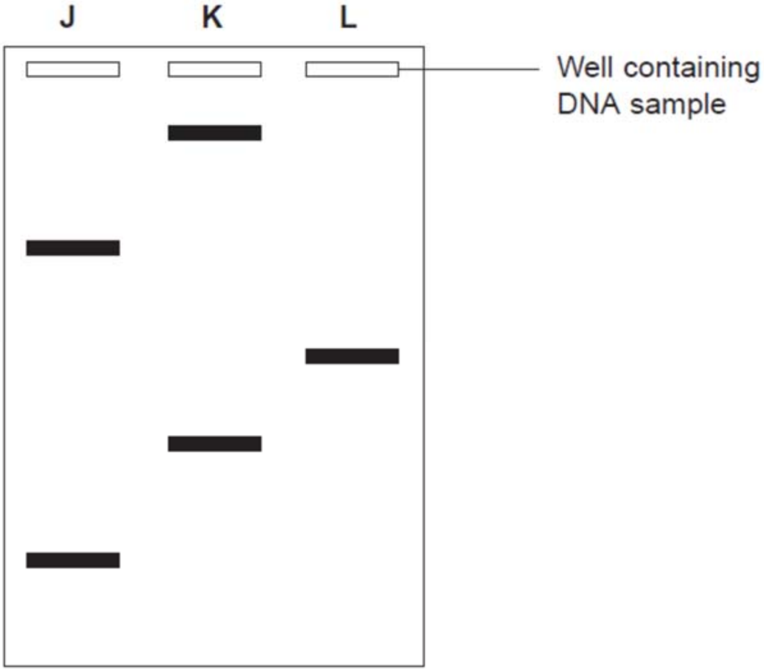

Fig. 2.3

(i) Only one of these people tested positive for Huntington's disease. Explain which person this is.

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(ii) Fig. 2.3 only shows part of the gel. Suggest how the scientists found the number of CAG repeats in the bands on the gel.

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(iii) Two bands are usually seen for each person tested. Explain why only one band was seen for Person **L**.

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[Total: 10]

3 There have been concerns that climate change and global warming could have serious consequences for the survival of the polar bears in the Arctic region. Whilst the actual impact of climate change on polar bear survival is not yet clear, there have been evidence that this could be detrimental to the polar bears in the long term.

Polar bears and their prey have evolved to live in the extreme conditions of the Arctic. Polar bears and seals depend on sea-ice for foraging, resting, and reproduction. The Arctic ecosystem was and continues to be shaped by climate.

There have been a number of climate change scenarios based on modelling of climate trend data in the Arctic. The more extreme models suggest that the Arctic basin will lose its ice within 50 years. Other models have suggested that ice thickness has decreased by 40% during the past 30 years and the average annual ice coverage in the polar region has diminished substantially, with an average annual reduction of over 1 million square kilometers.

Researchers have studied the polar bear population demographics in the Western Hudson Bay area of the United States since 1981. This region is the southern-most extent of the range of the polar bear. They have noticed that the sea ice breakup has been occurring earlier. The earlier breakup has been related to the poorer condition of polar bears and there is a correlation between the earlier breakup and a pattern of warming air temperatures during the spring between 1950 and 1990. It appears that earlier breakup caused by warmer temperatures has resulted in declines in physical and reproductive conditions of polar bears in this area.

Passage adapted from essay by:
Schliebe, S.L. – 'What has been happening to polar bears in recent decades?"
National Oceanic and Atmospheric Administration webite:
https://www.pmel.noaa.gov/arctic-zone/essay_schliebe.html

(a) Explain the term 'global warming'.

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For Examiner's Use

(b) Polar bears (*Ursus maritimus*) are thought to have evolved from brown bears or grizzly bears (*Ursus arctos*). This period coincided with the melting and retreat of the glaciers northward. Fig. 3.1 shows the divergence times of different bears with time in millions of years.

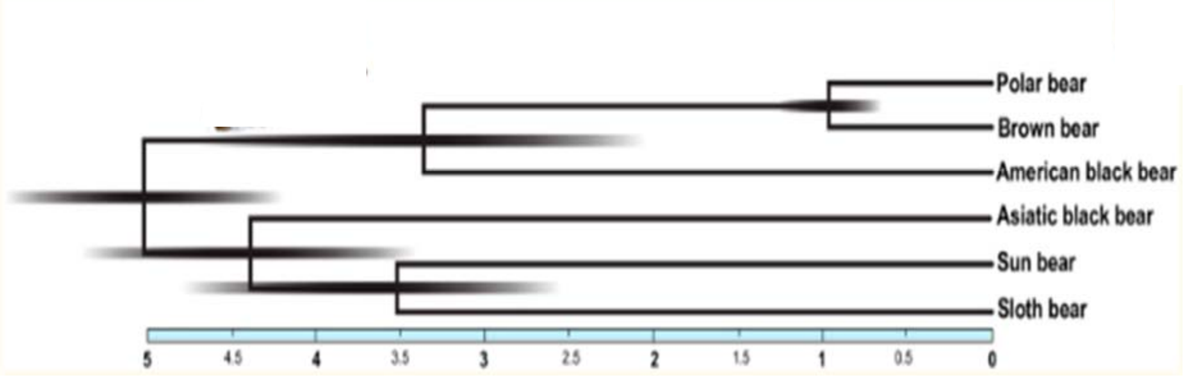

Fig. 3.1

(i) State what is meant by the term, *biological species.*

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(ii) Explain how the ancestors of the polar bear could have diverged from the brown bear about 1 million years ago.

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(iii) In recent years, there have been a discovery of hybrids between polar bears and brown bears (grizzly bears) called 'pizzlies' or 'grolars'. Using information from Fig. 3.1, suggest a possible reason why these hybrids are possible.

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(iv) Suggest how climate change can result in the polar bears and the brown bears (grizzly bears) interacting and interbreeding to produce these hybrids.

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(c) A student who is doing a project on the effects of climate change on the possible evolution of animals in the Arctic had read about the current trends in climate changes in the Arctic. He thinks that this could lead to the rapid deterioration of the polar bear population in areas like the Western Hudson Bay region.

This student wrote in the draft notes for his written report: "Polar bears may still survive in the face of climate change. For example, if one polar bear had the ability to adapt to warmer temperatures and could hunt effectively on land instead of on sea-ice, this polar bear would evolve into an animal that would be very well suited for the warmer temperatures of the Arctic region in the years to come."

Using your understanding of evolution and natural selection, evaluate the student's claim and justify with appropriate reasons.

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[Total: 10]

For Examiner's Use

Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

4 **(a)** Discuss the importance of ions in Biology. [12]

(b) Insulin and glucagon help to regulate blood glucose concentration. Disruption of this homeostatic mechanism can lead to Type 2 diabetes.

Describe the mode of action of insulin in regulation of blood glucose concentration and explain how disruption of the homeostatic mechanism leads to Type 2 diabetes. [13]

5 **(a)** Carbon dioxide may affect organisms directly or indirectly. Describe and explain these effects. [12]

(b) Discuss the control of processes in cells and the importance of these controls. [13]

- END OF PAPER -